Summary in Graph

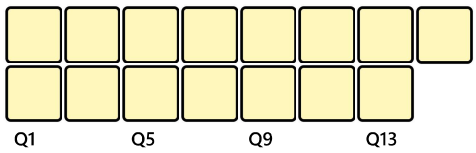
Exam Summary (GO Classes DA Test Series 2026 | Calculus and Optimization | Differentiability, maxima and minima | Topic Wise Test 2)

Qs. Attempted:	**0** (0 + 0)	**Correct Marks:**	**0** (0 + 0)
Correct Attempts:	**0** (0 + 0)	**Penalty Marks:**	**0** (0 + 0)
Incorrect Attempts:	**0** (0 + 0)	**Resultant Marks:**	**0** (0 + 0)

Total Questions:	**15** (10 + 5)
Total Marks:	**20** (10 + 10)
Exam Duration:	**45 Minutes**
Time Taken:	**0 Minutes**

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (61) out of 613 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Technical

Q #1 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

Let $f(x) = Ax^B \ln(x)$, where A and B are unspecified constants. Suppose that $\left(e^5, 10\right)$ is a point of local extremum for $f(x)$. Calculate the values of A and B.

A. $A = 2e, B = -\frac{1}{5}$
B. $A = 2e, B = \frac{1}{5}$
C. $A = e, B = -5$

D. $A = e, B = 5$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 02sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

If a and b are positive numbers, what is the maximum value of $f(x) = x^a(1-x)^b$ for $0 \leq x \leq 1$?

A. $\frac{a^a b^b}{(a+b)^{(a+b)}}$

B. $\frac{(a+b)^{(a+b)}}{a^a b^b}$

C. $\frac{a}{a+b}$

D. $\frac{(a+b)^{(a+b)}}{a^s b^\rho}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

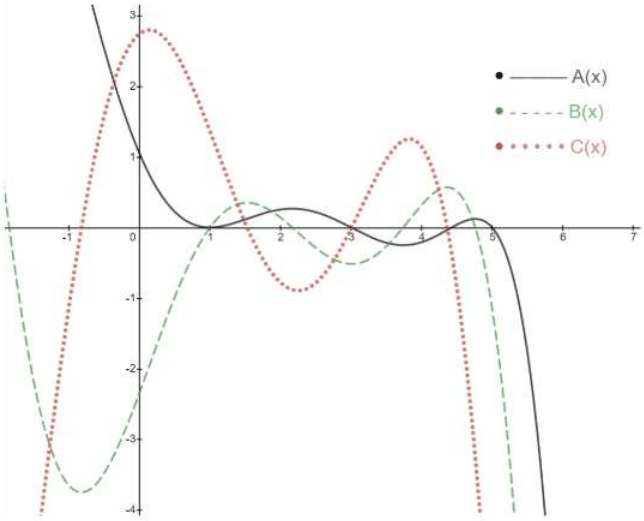
Consider the function $f(x) = (x-5)(x+10)^2 = x^3 + 15x^2 - 500$.
Find where f is concave up and find where f is concave down.
Which of the following statements is correct about the function $f(x)$?

A. f is concave down on $[-5, \infty)$, concave up on $(-\infty, -5]$.
B. f is concave down on $(-\infty, -5]$, concave up on $[-5, \infty)$.
C. f is concave down on $(-\infty, 5]$, concave up on $[5, \infty)$.
D. f is concave down on $[5, \infty)$, concave up on $(-\infty, 5]$.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

The figure below shows the graphs of f, f', and f''. Identify which graph is that of f''.



A. $f(x) = C(x), \quad f'(x) = A(x), \quad f''(x) = B(x)$
B. $f(x) = B(x), \quad f'(x) = C(x), \quad f''(x) = A(x)$
C. $f(x) = A(x), \quad f'(x) = B(x), \quad f''(x) = C(x)$
D. $f(x) = C(x), \quad f'(x) = B(x), \quad f''(x) = A(x)$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Select Type | Award: 1 | Penalty: 0 | Calculus

. Suppose f is differentiable everywhere. Which of the following formulas are equal to $f'(a)$, for every a ?

A. $\lim_{h \to 0} \frac{f(h)-f(a)}{h-a}$

B. $\lim_{x \to a} \frac{f(x)-f(a)}{x-a}$

C. $\lim_{h \to 0} \frac{f(a+h)-f(h)}{a}$

D. $\lim_{h \to 0} \frac{f(a+h)-f(a)}{h}$

Your Answer: | Correct Answer: B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Select Type | Award: 1 | Penalty: 0 | Calculus

Let $f(x)$ be continuous on $[1, 5]$ and differentiable on $(1, 5)$. Suppose that $f'(x) > 3$ for all x, and $f(1) = 1$. According to the Mean Value Theorem (MVT), what can be concluded about $f(5)$?

A. $f(5) > 13$
B. $f(5) < 13$
C. $f(5) = 13$
D. $f(5) \geq 12$

Your Answer: | Correct Answer: A;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

Which of the following completes the statement of the Mean Value Theorem applied to the function $f(x) = \frac{e^x}{x}$ on the interval $[1, 3]$?
"There is a point c in the interval $(1, 3)$ such that ..."

A. $\frac{xe^x - e^x}{x^2} = \frac{\frac{1}{3}e^c - e}{2}$

B. $\frac{ce^c - e^c}{c^2} = \frac{e^3}{9}$

C. $\frac{e^c}{c} = \frac{\frac{1}{3}e^3 - e}{3-1}$

D. $\frac{e^c(c-1)}{c^2} = \frac{e^3}{6} - \frac{e}{2}$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

If f is a continuous function on the closed interval $[a, b]$, which of the following must be true?

A. There is a number c in the open interval (a, b) such that $f(c) = 0$.
B. There is a number c in the open interval (a, b) such that $f(a) < f(c) < f(b)$
C. There is a number c in the closed interval $[a, b]$ such that $f(c) \geq f(x)$ for all x in $[a, b]$.
D. There is a number c in the open interval (a, b) such that $f'(c) = 0$.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Numerical Type | Award: 1 | Penalty: 0 | Calculus

Suppose that f is continuous and differentiable on the interval $[1, 6]$. Also suppose that $f(1) = -8$ and $f'(x) \leq 4$ for all x in the interval $[1, 6]$. What is the largest possible value for $f(6)$?

Your Answer: | Correct Answer: 12 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

Let $f(x)$ be a smooth function with exactly two inflection points at $x = -1$ and $x = 2$, and a local maximum at $x = 0$. Which of the following must be true?

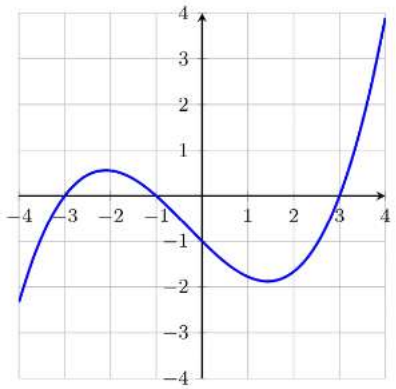
A. $f''(x) \geq 0$ on $(-1, 2)$
B. $f''(x) \leq 0$ on $(-1, 2)$
C. $f(x) \geq 0$ on $(-1, 2)$
D. $f(x) \leq 0$ on $(-1, 2)$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

Consider the function $y = g(x)$.

Consider the function $y = g(x)$.



Using the graph of $y = g(x)$, determine the sign of $g''(-3)$ and $h'(x)$ for $h(x) = [g(x)]^2$.

A. $g''(-3) > 0$ and $h'(-4) > 0$
B. $g''(-3) < 0$ and $h'(-4) > 0$
C. $g''(-3) > 0$ and $h'(-4) < 0$
D. $g''(-3) < 0$ and $h'(-4) < 0$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Select Type | Award: 2 | Penalty: 0 | Calculus

Consider the table below:

x	0	1	2	3	4
$f(x)$	0.5	2	2.5	0	-4
$f'(x)$	1.5	0.5	-1	-3	-3.5

Suppose $l(x)$ is a linear function of x, with $l(4) = 0$, and $l'(4) < f'(4)$. Which of the following statements about $l(x)$ is true? (Mark all that apply)

A. $l(x) > 0$ for $x > 4$.
B. $l(x) < 0$ for $x > 4$.
C. $l(x)$ is increasing for all x.
D. $l(x)$ is decreasing for all x.

Your Answer: | Correct Answer: B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

$S1$: If $f : (a, b) \to \mathbb{R}$ is a real-valued function differentiable at $p \in (a, b)$, then

$$\lim_{h \to 0} \frac{f(p + h) - f(p - h)}{2h} = f'(p)$$

$S2$: If a differentiable function $f : [a, b] \to \mathbb{R}$ is not one-to-one, i.e., there exist points $p, q \in [a, b]$ with $p \neq q$ such that $f(p) = f(q)$. By the Mean Value Theorem, there exists a point t between p and q such that

$f'(t) = 0$.

Which of the following option is correct?

A. Statement $\mathrm{S1}$ is true, and statement $\mathrm{S2}$ is false.
B. Statement $\mathrm{S1}$ is false, and statement $\mathrm{S2}$ is true.
C. Both statements $S1$ and $S2$ are true.
D. Both statements $S1$ and $S2$ are false.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Select Type | Award: 2 | Penalty: 0 | Calculus

Which of the following statements is/are NOT CORRECT?

A. If f is concave up on $[0, 1]$ and concave down on $[1, 2]$ then 1 is an inflection points.
B. If a function f is differentiable on $[-1, 1]$, then there is a point x in that interval where $f'(x) = 0$.
C. If $f''(-2) > 0$ then f is concave up at $x = -2$.
D. An inflection point is a point, where the function $f'(x)$ changes sign.

Your Answer: | Correct Answer: B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

Let $f(x)$ be de defined for all x. Suppose $f(x)$ is continuous and differentiable whenever $x \neq 0$ and satisfies

$$f(0) = 1 \lim_{x \to 0^+} f(x) = 1$$
$$\lim_{x \to 0^-} f(x) = a \lim_{x \to 0^+} f'(x) = b$$

Under what conditions on the constants a and b does $f(0)$ guarantee to be a local maximum?

A. $a < 1$ and $b > 0$
B. $a < 1$ and $b < 0$
C. $a > 1$ and $b < 0$
D. $a > 1$ and $b > 0$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

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